Import Data Using Transform Maps (Spreadsheet)

Phase 3 - Project Design

1. Design Overview

The **Project Design Phase** defines how the spreadsheet data import solution will be structured and implemented in ServiceNow.

The main objective is to design a process that can take data from a spreadsheet, load it into an Import Set, transform and validate the data using a Transform Map, and finally insert or update records in the target table.

The overall design is:

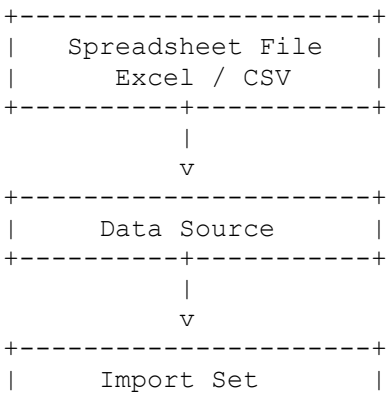
Spreadsheet → Data Source → Import Set → Import Set Table → Transform Map → Target Table

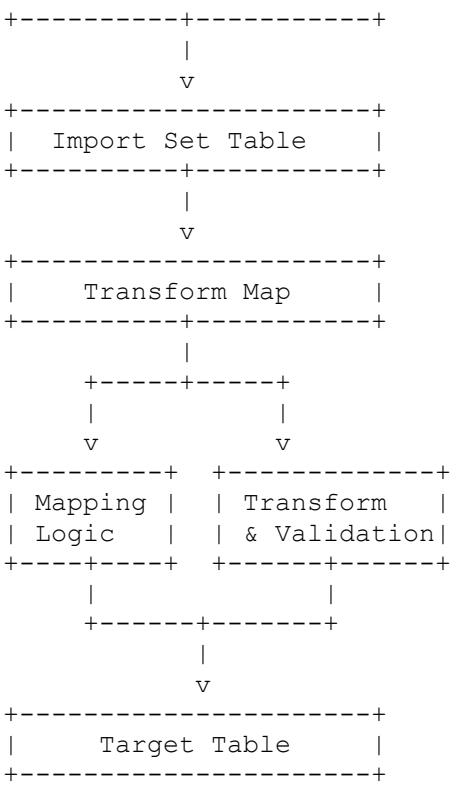
2. System Architecture Design

The solution consists of the following major components:

1. **Source Spreadsheet**
2. **Data Source**
3. **Import Set**
4. **Import Set Table**
5. **Transform Map**
6. **Field Mapping**
7. **Transformation Logic**
8. **Validation and Error Handling**
9. **Target Table**

Architecture Flow





3. Source Data Design

The source spreadsheet should follow a predefined structure.

For example:

Employee ID	Employee Name	Email	Department	Joining Date
E001	Arun	arun@example.com	IT	10-01-2025
E002	Kumar	kumar@example.com	HR	15-02-2025

The spreadsheet should contain:

- Proper column headers.
- Consistent data formats.
- Required fields.
- Unique identifiers where applicable.
- No unnecessary columns.
- Valid data values.

A standard spreadsheet template can be created so that users follow the same structure for every import.

4. Data Source Design

A **Data Source** will be designed in ServiceNow to define where the source data comes from.

The Data Source configuration will specify:

- Source file type.
- File format.
- Import method.
- Import Set Table.
- File-related configuration.

For this project, the spreadsheet will act as the external source of data.

5. Import Set Design

The **Import Set** is designed as an intermediate layer between the spreadsheet and the final target table.

Its purpose is to:

- Receive spreadsheet data.
- Store the imported source records temporarily.
- Separate raw source data from production data.
- Provide a staging area before transformation.

The Import Set allows the data to be reviewed and transformed before it reaches the target table.

6. Import Set Table Design

An **Import Set Table** will be designed to temporarily store the spreadsheet data.

The fields in the Import Set Table should correspond to the columns in the spreadsheet.

Example:

Import Set Field	Data Type
Employee ID	String
Employee Name	String
Email	String
Department	String
Joining Date	Date

The Import Set Table acts as the **source table** for the Transform Map.

7. Transform Map Design

The **Transform Map** is the main component of the project.

It defines how data should move from the Import Set Table to the target ServiceNow table.

The Transform Map will contain:

- Source table.
- Target table.
- Field mappings.
- Coalesce configuration.
- Transformation logic.
- Optional Transform Scripts.
- Related transformation settings.

Example

Source: Import Set Table

Target: Employee Table

Source Field	Target Field
Employee ID	Employee ID
Employee Name	Name
Email	Email
Department	Department
Joining Date	Date of Joining

8. Field Mapping Design

Each source field will be mapped to the corresponding target field.

For example:

```
Employee ID      → Employee ID
Employee Name    → Name
Email            → Email
Department       → Department
Joining Date     → Date of Joining
```

The mapping design should ensure:

- Correct source-to-target relationship.

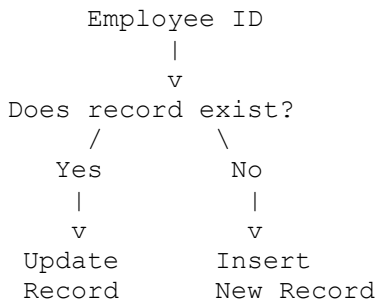
- Compatible data types.
- Mandatory target fields are populated.
- Unnecessary fields are ignored.
- Reference fields are handled properly.

9. Coalesce Design

The project will use **Coalesce** to identify existing records.

For example, **Employee ID** can be used as the unique identifier.

The design logic is:



This design helps prevent unnecessary duplicate records.

10. Data Transformation Design

Transformation logic will be designed for fields where the source format differs from the target format.

Possible transformation requirements include:

- Date format conversion.
- Text standardization.
- Removing leading/trailing spaces.
- Converting values to the required format.
- Mapping source values to reference records.
- Handling null or empty values.

For simple requirements, field mapping can be sufficient.

For complex requirements, **Transform Scripts** can be incorporated into the design.

11. Data Validation Design

Validation will be incorporated into the transformation process.

The design should verify:

- Mandatory fields are populated.
- Employee ID is available.
- Email format is valid.
- Department value is valid.
- Date values are in the expected format.
- Duplicate records are properly handled.

Invalid data should be identified rather than creating incorrect target records.

12. Error Handling Design

The system should provide a mechanism for identifying failed records.

Possible errors include:

- Missing mandatory fields.
- Invalid field values.
- Invalid reference values.
- Incorrect date formats.
- Mapping failures.
- Transformation errors.

The design should allow users to review the failed records, correct the source spreadsheet, and perform the import again.

13. Target Table Design

The target table will contain the final transformed records.

Example target table:

Field	Type	Requirement
Employee ID	String	Mandatory / Unique
Name	String	Mandatory
Email	Email/String	Mandatory
Department	Reference/String	Required
Phone	String	Optional
Date of Joining	Date	Required

14. Security Design

The import process should follow ServiceNow access controls.

The design should ensure:

- Only authorized users can upload/import data.
- Users have appropriate access to the Import Set.
- Users have appropriate access to the target table.
- Sensitive information is protected.
- Imported records follow applicable access controls.

15. User Interface Design

The user interface should provide a simple process for importing the spreadsheet.

The user workflow can be:

1. Open the data import option.
2. Select the spreadsheet.
3. Upload the file.
4. Select the appropriate Import Set/Data Source.
5. Run the import.
6. Execute the Transform Map.
7. Review import results.
8. Verify the target records.

The interface should make the import process understandable even for users with limited technical knowledge.

16. Database/Data Flow Design

The data flow can be represented as:

```
Spreadsheet
     |
     v
Data Source
     |
     v
Import Set
     |
     v
Import Set Table
     |
     v
Transform Map
     |
```

```
      +----> Field Mapping
      |
      +----> Data Transformation
      |
      +----> Validation
      |
      +----> Coalesce Check
      |
      v
Target Table
      |
      +----> New Record
      |
      +----> Existing Record Update
```

17. Testing Design

The project design should include different test scenarios.

Test Scenario	Expected Result
Valid spreadsheet	All valid records imported
Missing mandatory field	Record identified as failed/invalid
Existing Employee ID	Existing record updated
New Employee ID	New record created
Invalid email	Validation/error handling triggered
Invalid date	Transformation/import error identified
Duplicate data	Duplicate handling applied
Empty spreadsheet	Appropriate validation/error shown

18. Design Considerations

The following points should be considered during implementation:

- Spreadsheet format should remain consistent.
- Source and target field names should be clearly documented.
- Unique identifiers should be defined.
- Mandatory fields should be identified.
- Transformation rules should be documented.
- Error handling should be tested.
- Import performance should be considered for large datasets.
- Security permissions should be configured appropriately.

19. Final Project Design

The final design for the project can be summarized as:

The project is designed to import spreadsheet data into ServiceNow using an Import Set and Transform Map architecture. The spreadsheet acts as the source, while the Import Set Table temporarily stores the source data. The Transform Map performs field mapping, data transformation, validation, and duplicate handling. Based on the transformation results, new records are inserted or existing records are updated in the target table. Error handling and security controls are included to ensure a reliable and controlled data import process.

Complete Design Flow

Spreadsheet
↓
Data Source
↓
Import Set
↓
Import Set Table
↓
Transform Map
↓
Field Mapping
↓
Transformation & Validation
↓
Coalesce / Duplicate Check
↓
Insert or Update Target Record
↓
Import Verification & Error Review